

Reverse physics and pure-Weyl gravity

Executive summary for physicists

By GPT-5.6.sol and Asger Alstrup Palm — 17 August 2026

Research context. Asger Alstrup Palm, a computer scientist and Honorary Professor at DTU Compute, orchestrates the programme and is the accountable human contact. AI systems perform substantial derivation, programming, literature work, adversarial review, and drafting. The work should be judged from its arguments, claim boundaries, and reproducible evidence—not accepted or rejected because of its unusual authorship.

Abstract

This paper develops a common framework for two questions. The first is whether pure Weyl gravity and related fourth-order theories survive classical, nonlinear, causal, quantum, and empirical tests. The second is which parts of those tests depend on physical postulates, which depend on the mathematical carrier, and which depend on logic, infinity, or existence principles. The framework treats a physical conclusion as a typed judgement

$$L + S + M + \text{Enc}(P) \vdash O,$$

where logic L , set or existence theory S , mathematical carrier M , and encoded physical postulates P are kept separate from the obligation O . This decomposition prevents a finite calculation, Euclidean measure, indefinite completion, cohomology class, or fitted curve from acquiring a stronger physical interpretation merely by notation.

Five case studies make the method concrete:

- fourth-order completion, compact causal reduction, and a defective Schwarzschild response distinguish local modes from globally admissible classical directions;
- an independently imported pure-Weyl BV complex supports typed nonlinear Green compatibility and a full 386-row BRST–Hadamard two-point **pseudo-state pair**, while positivity and a physical graviton Hilbert space remain separate obligations;
- a bounded NGC 3198 comparison gives Mannheim conformal gravity the smaller unweighted residual, but GR plus an NFW halo the better weighted fit and the only passing random-error gate among the three galactic models tested;
- an explicit Bateman–Turok Euclidean torus family refutes a tempting deterministic all-field stability bound at the free spectral scale;
- eight end-to-end theory passports and nine programme prototypes expose where mathematical pieces exist but the joins to predictions or data are absent.

These results neither select an unconventional theory nor reduce the subject to a generic ghost objection. They support a more modest thesis: assumptions, carriers, proofs, interfaces, and empirical tests can be separated into reviewable obligations, and doing so exposes both genuine theorems and missing bridges.

The obligation ladder

The basic unit of analysis is not a theory name but an obligation. A proposed theory must specify what counts as a state, how states evolve, which quantities are observable, how gauge redundancy is removed, what probability means, how local constructions become global or continuum objects, and how a numerical prediction meets data. Evidence at one rung does not automatically climb to the next.

Algebraic and classical obligations

At the algebraic end, selected fourth-order systems admit positive, Krein, vacuum, and Jordan completions, while explicit interactions obstruct some analytic positive continuations. In pure-Weyl gravity, the conformal-cylinder BV–BFV construction supplies residual cohomology and causal transport in a declared polarization.

Compact axial and polar sectors also contain additional nonradical directions whose action-derived pairing does not obey the shortcut “Einstein positive, additional negative.”

The quantum programme imports this classical complex through a content-addressed Gate-A snapshot. Nilpotency, contraction, chain-map, cyclicity, residual cohomology, and pairing checks certify the import in its stated scope. They do not turn the centered residual classes $[W_+^2]$ and $[W_-^2]$ into one-particle gravitons; those classes remain deformations or interaction vertices.

Nonlinear and quantum obligations

Finite-harmonic nonlinear continuation and several obstruction branches are known without a complete bounded nonlinear cone. At one loop, strict pure Weyl has a nonzero local Euclidean BV anomaly. A formal shifting compensator repairs the tested identity only after enlarging the field content and counterterm space; it is therefore a result about a changed theory.

On the Lorentzian side, typed q_2/q_3 Green compatibility leads to a full-complex BRST–Hadamard two-point pair on the 386-row off-shell carrier. It is a pseudo-state: the construction supplies the declared bisolution, graded CCR, wavefront, Ward, reality, and stationarity conditions without supplying positivity on physical cohomology. Consequently it does not establish a physical Hilbert space, renormalized time-ordered products, causal pAQFT, a Lorentzian QME, scattering, or unitarity.

Propagation and empirical obligations

The axial Schwarzschild analysis proves endpoint non-selection, a non-split repeated Regge–Wheeler extension, a defective $\ell = 2$ quasinormal mode, and a nonzero second-order pole of a compactly cut-off radial Green operator. The pole has a retarded mode-reduced parent, but a mode-reduced meromorphic bridge is not a complete global metric or Bondi waveform.

Empirical gates are likewise scoped. Standard GR passes the declared Cassini test, and GR+NFW passes the bounded NGC 3198 random-error gate. On that same galaxy protocol, Mannheim conformal gravity has the smaller unweighted residual but fails the uncertainty-weighted gate. None of these statements is a population-level comparison with astrophysical systematics.

Foundation and composition obligations

The reverse-physics atlas crosses six foundational regimes, six carrier families, and sixteen physical obligations, giving 576 navigational coordinates. Exact finite, constructive, weak-foundation, Hilbert, Krein, algebraic, smooth, and internal constructions can therefore be compared without being identified. A populated cell records scoped evidence; it is not a complete theory.

The nine programme prototypes are coverage envelopes, not rankings. The eight theory passports ask whether foundations, state space, dynamics, observables, predictions, and empirical tests are joined for one route. Two Lean/Physlib proof passports independently replay finite identities, but neither formal proof supplies a missing physical premise or promotes an atlas grade.

A type system for scientific claims

Every quantum result carries one or more dependency tags:

LOCAL-ALGEBRAIC
EUCLIDEAN-SPECTRAL
REDUCED-MODE
LORENTZIAN-CAUSAL

A reduced-mode calculation is not evidence for a Lorentzian-causal theorem. A Euclidean determinant is not a Lorentzian state. A cohomology class is not a particle. A finite exact model is not a continuum limit.

Quantum lifecycle states are also distinct:

CLASSIFIED
COEFFICIENT_COMPUTED
QME_RESTORED
RESIDUAL_TRANSFERRED
LORENTZIAN_CERTIFIED

The programme deliberately permits later stages to remain visible when an earlier interface is open. That is diagnostic information, not permission to compose around the missing step.

Five case studies

1. Fourth-order completion and the ghost test

Papers 01–06 distinguish a positive finite-dimensional metric, a Krein real form, vacuum covariance, field representation, Fock sector, the equal-frequency Jordan limit, and interaction stability. These objects are related but not interchangeable. A local negative sign does not by itself prove a negative probability particle; conversely, a positive metric on a free block does not guarantee a compatible interacting theory.

Papers 07–18 carry this separation into pure-Weyl gauge theory, compact model universes, Schwarzschild end-points, and black-hole thermodynamics. The stable Phase-1 conclusion is a classification: additional classical directions survive some reductions, while global balance, resonance, stability, and quantum tests obstruct others. No tested branch passes every level.

In axial Schwarzschild $\ell = 2$, ordinary future-horizon and one-sided radiation conditions do not select the Einstein subsector. A repeated spin-two Regge–Wheeler pair forms a non-split self-extension. At one validated damped frequency it has a length-two root chain and a rank-one second-order cut-off Green pole. The local contour contains a $te^{i\omega_n t}$ term, but no complete causal detector waveform has been proved.

2. Classical authority, quantum anomaly, and Hadamard data

The classical BV–BFV complex is the authoritative input to the quantum programme. Gate A imports a content-addressed snapshot rather than rebuilding a competing complex in the quantum tree. The snapshot passes the declared algebraic, homological, cyclic, residual, and pairing checks.

On the imported complex, typed q_2/q_3 compatibility with advanced and retarded Green homotopies supports a BRST–Hadamard two-point pair on the full 386-row carrier. The word **pseudo-state** is load-bearing. The pair is indefinite and does not establish positivity on physical cohomology, a unique zero-mode policy, a Feynman propagator, renormalized products, causal pAQFT, QME restoration, particle space, scattering, or unitarity.

This Lorentzian frontier is logically separate from Paper 12’s local anomaly classification. Strict pure Weyl is locally QME-obstructed at one loop in the declared Euclidean quotient. A formal compensator extension restores a changed local theory at that order. Neither result is a positive Lorentzian quantum theory.

3. Mannheim conformal gravity and bounded data contact

Papers 18–19 isolate what the static Mannheim–Kazanas explanation assumes. The linear potential is forced by the radial biharmonic equation, but its physical interpretation depends on matter, conformal frame, source moments, and boundary data. In the simplest constant-scalar frame the linear term is removed and massive particles follow the conformally rescaled geometry.

The common NGC 3198 protocol compares 39 points and quoted random errors:

- Newtonian baryons: RMS 23.896 km/s, $\chi_\nu^2 = 128.722$ — fail.
- GR+NFW: RMS 5.148 km/s, $\chi_\nu^2 = 0.965$ — pass.
- Mannheim: RMS 4.694 km/s, $\chi_\nu^2 = 3.202$ — fail.

Mannheim therefore has the smallest unweighted RMS, while NFW has the better uncertainty-weighted score and AICc. This is one bounded galaxy protocol. It omits major systematics and provides neither population-level model selection nor a universal refutation.

4. Bateman–Turok and the failure of an all-field shortcut

The Bateman–Turok positive Euclidean lattice and hidden ghost-parity proposal raises a natural deterministic question: can the complete action gradient be bounded below at the free bilaplacian scale for every positive field?

Paper 22 answers **no**. On $L_n = n^{24}$, an exactly mean-compensated critical source is inverted with the periodic graph Green operator, lifted above its minimum, and given a small four-way coupling. The resulting fields are positive, nonseparable, and polynomial-contrast. Their action stays bounded above and away from zero, while

$$\frac{Q_n}{\omega_{L_n}^2} = O(n^{-2}) \rightarrow 0.$$

This refutes one deterministic all-field scaled-PL architecture. It does not refute the full Bateman–Turok programme. A successful route could still use Gibbs typicality, the actual interacting Witten or H^{-1} structure, or a different reconstruction argument.

5. Reverse foundations and theory assemblies

Papers 20–21 and the interactive atlas ask whether changing the mathematical assumptions changes what physics can be built. The axes include classical, choice-weakened, constructive, internal/topos, and finite regimes; finite, Hilbert, Krein, algebraic, smooth/distributional, and localic carriers; and sixteen physical obligations from states and dynamics through renormalization, reconstruction, and empirical adequacy.

The atlas contains nine programme prototypes. The mainstream GR/QFT reference has direct records for 16/16 jobs, while the Mannheim and pure-Weyl prototypes each show 15/16 and other programmes less. Those numbers are coverage envelopes, not rankings. A prototype may select different models and carriers for different jobs; no result follows until their interfaces are proved.

Eight theory passports instead follow six stage functions—foundations, state space, dynamics, observable, prediction, and empirical benchmark. Four reach bounded data: GR/Cassini and GR+NFW/NGC 3198 pass their declared gates; Newtonian baryons and Mannheim/NGC 3198 fail theirs. Bateman–Turok, the free Krein mode, the constructive coded wave, and the pure-Weyl BV route have not yet reached an empirical test.

How to review the work

Start with the [Reverse Physics Atlas](#):

1. **Dimensions guide** explains every axis in non-specialist language.
2. **Theory profiles** shows obligation coverage without pretending it is empirical validity.
3. **Assemblies** separates bounded model tests, research programmes, and interface/calibration ledgers.
4. **Theory journeys** exposes each end-to-end passport and its first blocker.
5. **Weyl BV routes** shows the eleven-gate causal and quantum frontier.
6. **Evidence** exposes certificates, literature records, and the two Lean/Physlib proof passports.

For a technical claim, follow

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claim
  → producer
  → machine-readable certificate
  → independent verifier
  → adversarial or mutation test
  → paper with explicit non-claims
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Re-running a producer is reproduction, not independent verification. Exact and numerical evidence remain different types. A timeout or skip is never a pass.

Supporting arguments

The completion and interaction analysis is developed in Papers 01–06; the residual and causal pure-Weyl construction in Papers 07–08; the compact laboratory, anomaly, and nonlinear analyses in Papers 09–13; and the black-hole architecture in Papers 14–18. Four papers isolate the cross-cutting arguments used here:

- [Paper 19](#) separates the radial Mannheim solution from its matter, frame, boundary, and empirical premises.
- [Paper 20](#) asks whether a gauge algebra is input, derived structure, or representation-dependent.
- [Paper 21](#) develops the reverse-foundations framework, literature synthesis, atlas, and evidence appendices.
- [Paper 22](#) proves the compensated Green-tail counterfamily for the deterministic BT torus bound.

Consequences for theory construction

The examples expose three recurring forks. First, an indefinite causal two-point object must either become positive after physical reduction or yield an exact positivity obstruction; neither outcome follows from Hadamard form alone. Second, a portfolio of individually supported ingredients must acquire typed interface theorems before it becomes an end-to-end theory. Third, a bounded empirical success or failure must survive systematics, held-out cases, and physically constrained priors before it bears on a research programme as a whole.

The same logic applies to foundation changes. Avoiding Choice in one finite or constructive representation establishes a local economy of assumptions, not a weakest-foundation theorem for physics. Conversely, discovering that a proof uses Hilbert completion, a smooth continuum, or classical existence does not show that nature requires it. The scientific question is whether the relevant obligation can be reconstructed on another carrier while preserving its operational meaning.

Bottom line

The programme has not shown that Weyl gravity or Bateman–Turok physics is a viable description of nature. It has also not reduced their questions to one generic “ghost” objection.

Its strongest durable result is methodological and technical:

Physical postulates, mathematical carriers, logical strength, causal and quantum lifecycle states, interface proofs, and empirical gates can be separated—and each separation exposes a concrete theorem or missing bridge that experts can inspect.

That makes the repository a research instrument, not a verdict machine.